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PROBLEMS SUBMIT STATUS STANDINGS CUSTOM TEST

E. XOR and Favorite Number

time limit per test: 4 seconds
memory limit per test: 256 megabytes

Bob has a favorite number k and a_i of length n . Now he asks you to answer m queries. Each query is given by a pair l_i and r_i and asks you to count the number of pairs of integers i and j , such that $l \leq i \leq j \leq r$ and the xor of the numbers $a_i, a_{i+1}, \dots, a_j$ is equal to k .

Input

The first line of the input contains integers n , m and k ($1 \leq n, m \leq 100\,000$, $0 \leq k \leq 1\,000\,000$) — the length of the array, the number of queries and Bob's favorite number respectively.

The second line contains n integers a_i ($0 \leq a_i \leq 1\,000\,000$) — Bob's array.

Then m lines follow. The i -th line contains integers l_i and r_i ($1 \leq l_i \leq r_i \leq n$) — the parameters of the i -th query.

Output

Print m lines, answer the queries in the order they appear in the input.

Examples

input	Copy
6 2 3 1 2 1 1 0 3 1 6 3 5	
output	Copy
7 0	

input	Copy
5 3 1 1 1 1 1 1 1 5 2 4 1 3	
output	Copy
9 4 4	

Note

In the first sample the suitable pairs of i and j for the first query are: (1, 2), (1, 4), (1, 5), (2, 3), (3, 6), (5, 6), (6, 6). Not a single of these pairs is suitable for the second query.

In the second sample xor equals 1 for all subarrays of an odd length.

Codeforces Round 340 (Div. 2)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363750954	Feb/21/2026 00:35	Accepted
363747704	Feb/20/2026 23:45	Accepted
363715484	Feb/20/2026 18:31	Accepted
363715109	Feb/20/2026 18:28	Accepted
363714985	Feb/20/2026 18:27	Accepted
363714116	Feb/20/2026 18:20	Accepted

→ Problem tags

data structures *2200

No tag edit access

→ Contest materials

Announcement (en)

Tutorial (en)